

SPEECHMASTER: CONFIDENCE-ROUTED SELF-SUPERVISED ASR WITH DISCRETE UNIT BUDGETING

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ABSTRACT

Self-supervised speech encoders are strong ASR front ends, but deploying the largest recognizer on every utterance wastes computation and obscures the accuracy/representation-cost tradeoff. We present SpeechMaster, a budget-aware ASR framework that composes heterogeneous self-supervised representations. A fast Wav2Vec2 CTC branch decodes every utterance, a label-free CTC risk score routes uncertain utterances to a stronger HuBERT branch, and a HuBERT unit auditor measures token rate and bitrate under discrete codebooks. On 1024 LibriSpeech dev-clean utterances, routing only 25% of utterances reduces WER from 2.94% to 2.40%, while full HuBERT decoding reaches 1.82%. On 1024 test-clean utterances, the same router improves WER from 3.61% to 3.04% at a 25% routing budget and to 2.60% at a 50% budget. An oracle route gives 1.45% dev WER and 1.92% test WER, showing complementary SSL-ASR errors.

Index Terms— ASR, self-supervised speech, uncertainty routing, HuBERT, Wav2Vec2

1. INTRODUCTION

Low-resource ASR benefits from self-supervised learning (SSL) because encoders such as Wav2Vec2, HuBERT, and WavLM learn phonetic and lexical structure from unlabeled audio. A common downstream design selects one SSL checkpoint, attaches or reuses a CTC decoder, and reports WER. This single-model view misses a practical question: many utterances are easy for a fast recognizer, while a smaller subset needs a stronger model. A useful low-resource system should therefore expose an accuracy/compute knob instead of committing to one encoder.

SpeechMaster treats SSL-ASR as a routed composition. It always runs a fast Wav2Vec2 branch, estimates utterance risk from CTC posterior geometry, and spends the HuBERT large branch only on utterances selected by a budget. In parallel, a discrete-unit auditor clusters HuBERT hidden states to quantify the bitrate needed by unit-based downstream variants. The contributions are: (1) a confidence-routed SSL-ASR framework, (2) a label-free CTC risk score with budgeted inference, (3) a discrete-unit bitrate auditor over HuBERT layers and codebook sizes, and (4) reproducible dev/test ex-

periments with WER, CER, RTF, token rate, bitrate, and routing ablations.

2. SPEECHMASTER

2.1. Two-Branch SSL-ASR

Given waveform x , a pretrained SSL encoder f_θ produces contextual hidden states $H = f_\theta(x)$. A CTC projection maps each frame to token posteriors:

$$p_\theta(y_t | x) = \text{softmax}(WH_t + b). \quad (1)$$

SpeechMaster uses Wav2Vec2 base as the fast branch F and HuBERT large as the strong branch S . Both are public SSL-ASR checkpoints; SpeechMaster itself adds no extra supervised training.

2.2. CTC Risk Router

Let q_t denote the fast-branch posterior and let $\mathcal{T}$ be non-blank argmax frames. SpeechMaster computes a risk score from posterior peakiness and entropy:

$$c(x) = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \max_k q_t(k) - \lambda \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} H(q_t), \quad (2)$$

where $\lambda = 0.03$. For routing budget B , utterances with the lowest $c(x)$ are decoded by S ; all others keep the fast hypothesis:

$$\hat{y}(x) = \begin{cases} S(x), & x \in \text{BottomB}(c), \\ F(x), & \text{otherwise.} \end{cases} \quad (3)$$

The same saved branch outputs are reused to ablate peak-only, entropy-only, duration, random, and oracle routers.

2.3. Discrete Unit Auditor

To connect ASR accuracy with representation compression, SpeechMaster extracts HuBERT hidden states from several layers and trains MiniBatchKMeans codebooks. For codebook size K and token rate r_{tok} , the nominal bitrate is

$$\text{bitrate} = r_{\text{tok}} \log_2 K. \quad (4)$$

We also collapse consecutive duplicate units, giving a compact token stream useful for text-to-unit or unit-to-ASR analyses.

Table 1. SpeechMaster routing on 1024 LibriSpeech dev-clean utterances.

Route	Routed utt.	WER	CER	RTF
0%	0	2.943	0.990	0.001
10%	102	2.668	0.875	0.001
25%	256	2.398	0.767	0.002
50%	512	2.044	0.654	0.002
75%	768	1.921	0.610	0.002
100%	1024	1.823	0.575	0.003
Oracle	-	1.450	0.520	-

Table 2. Single-model ASR and layer ablations on dev-clean.

System	Rep.	N	WER	CER	RTF
wav2vec2-base-960h	final	1024	2.943	0.990	0.001
wav2vec2-base-960h	last4	1024	3.327	1.111	0.002
wav2vec2-base-960h	midlast	1024	3.705	1.140	0.001
hubert-large-ls960-ft	final	1024	1.823	0.575	0.003
wav2vec2-large-960h-lv60-self	final	1024	1.636	0.532	0.003

3. EXPERIMENTAL SETUP

Experiments use LibriSpeech clean splits at 16 kHz. The main development suite uses 1024 dev-clean utterances; the final validation suite uses 1024 test-clean utterances. Unit analysis uses 512 dev-clean utterances. The fast branch is Wav2Vec2-Base-960h; the strong branch is HuBERT-Large-960h; a stronger reference model is Wav2Vec2-Large-LV60. HuBERT unit analysis uses HuBERT-Base. Exact checkpoint identifiers are listed in the released scripts. Metrics are WER, CER, and real-time factor (RTF) for ASR, plus token rate, deduplicated token rate, bitrate, and codebook size for discrete units.

4. RESULTS

4.1. Main Dev-Clean Results

Table 1 is the main SpeechMaster development result. At a 25% routing budget, WER drops from 2.94% to 2.40%, a relative reduction of 18.5%, while using substantially less strong-branch decoding than 100% HuBERT. At 50%, WER reaches 2.04%. The oracle route is lower than both branches on dev-clean, showing that fast and strong SSL recognizers make non-identical errors.

Table 2 reports single-model and layer ablations. The final Wav2Vec2 CTC layer is stronger than last-layer averaging or mid/final blending, indicating that the pretrained CTC head is calibrated to the final state. The large LV60 Wav2Vec2 model is the best single model on this dev slice, but the fast+HuBERT oracle still improves over it.

Table 3. SpeechMaster routing on 1024 LibriSpeech test-clean utterances.

Route	Routed utt.	WER	CER	RTF
0%	0	3.614	1.186	0.001
10%	102	3.398	1.093	0.001
25%	256	3.035	0.946	0.001
50%	512	2.602	0.797	0.002
75%	768	2.292	0.698	0.002
100%	1024	2.231	0.667	0.002
Oracle	-	1.920	0.658	-

Table 4. Selected HuBERT unit statistics on 512 dev-clean utterances.

Layer	K	Tok/s	Dedup tok/s	Dedup bit/s
-8	100	49.90	28.08	186.58
-8	1000	49.90	35.17	350.46
-4	100	49.90	27.14	180.34
-4	1000	49.90	29.74	296.36
-1	100	49.90	26.67	177.19
-1	1000	49.90	34.50	343.85

4.2. Test-Clean Verification

Table 3 verifies the same framework on test-clean. The absolute WERs are higher than dev-clean, but the monotonic routing behavior remains: 25% routing improves WER from 3.61% to 3.04%, and 50% routing reaches 2.60%. The oracle route reaches 1.92%, confirming remaining headroom for better risk prediction.

4.3. Router and Unit Ablations

Routing-score ablations show that risk-aware ordering is consistently better than random routing. On dev-clean at 25%, random routing gives 2.66% WER, duration gives 2.54%, entropy gives 2.31%, and SpeechMaster peak-entropy gives 2.40%; the oracle gives 1.45%. On test-clean at 50%, random gives 2.93%, duration gives 2.62%, peak-only gives 2.59%, and peak-entropy gives 2.60%. Thus the proposed score is robust, but the oracle gap indicates that a learned router is a promising extension.

The unit auditor exposes a compression tradeoff. Raw HuBERT assignments appear at about 49.9 tokens/s for all tested layers. Deduplication roughly halves the rate: for example, layer -4 with $K = 100$ yields 27.14 deduplicated tokens/s and 180.34 bit/s, while $K = 1000$ raises this to 29.74 tokens/s and 296.36 bit/s. Layer -8 keeps more rapidly changing units than layer -4 at the same codebook size, suggesting a higher acoustic-detail bitrate.

5. CONCLUSION

SpeechMaster combines fast and strong self-supervised ASR models through label-free CTC risk routing and audits the bitrate of HuBERT discrete units. Across dev-clean and test-clean, routing a limited fraction of utterances gives a smooth WER/RTF tradeoff, while oracle routing reveals substantial complementarity. The results support a system-level view of low-resource SSL-ASR: encoder choice, routing budget, and representation bitrate should be optimized jointly rather than treated as isolated design choices.

6. REFERENCES

7. REFERENCES

- [1] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, “Librispeech: An ASR corpus based on public domain audio books,” in *Proc. ICASSP*, 2015.
- [2] A. Baevski, H. Zhou, A. Mohamed, and M. Auli, “wav2vec 2.0: A framework for self-supervised learning of speech representations,” in *NeurIPS*, 2020.
- [3] W.-N. Hsu et al., “HuBERT: Self-supervised speech representation learning by masked prediction of hidden units,” *IEEE/ACM TASLP*, 2021.
- [4] S. Chen et al., “WavLM: Large-scale self-supervised pre-training for full stack speech processing,” *IEEE JSTSP*, 2022.
- [5] Jitsi, “JiWER: Word error rate and character error rate for automatic speech recognition,” <https://jitsi.github.io/jiwer/>.